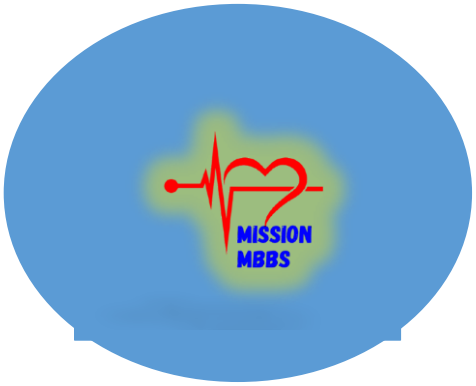
Mission MBBS TEAM
Your MISSION IS OUR MISSION

HEC USAT Pre Test



INSTRUCTIONS;
Duration of Test is 120 minutes (100 Questions).
No Negative Marking
Fill only one circle each question on given respective sheet .
Answer key will be displayed on your page Mission MBBS (Facebook) for MCQS discussion YouTube (Mission MBBS) .
Incase of any Confusion get help from Mission MBBS TEAM.

1.Which of the following organelles is responsible for protein synthesis?

a) Golgi apparatus
b) Mitochondria
c) Ribosomes
d) Lysosomes

2.The primary function of DNA is to:

a) Provide energy for cellular processes
b) Store and transmit genetic information
c) Act as a structural component of cell membranes
d) Catalyze biochemical reactions

3.Enzymes function by:

a) Increasing the activation energy of a reaction
b) Lowering the activation energy of a reaction
c) Adding energy to a reaction
d) Removing energy from a reaction

4.Cellular respiration is a process that:

a) Releases energy stored in food molecules
b) Stores energy in food molecules
c) Converts light energy into chemical energy
d) Breaks down water molecules

5.Viruses are considered non-living because they:

a) Lack a cellular structure
b) Cannot reproduce independently
c) Cannot metabolize on their own
d) All of the above

6.Bacteria are classified into different groups based on their:
a) Shape
b) Mode of nutrition
c) Cell wall composition
d) All of the above

7.Angiosperms are characterized by:

a) Naked seeds
b) Vascular tissue
c) Flowers and fruits
d) Motile gametes

8.Mammals are distinguished from other vertebrates by:

a) Hair or fur
b) Mammary glands
c) Endothermy
d) All of the above

9.The heart pumps blood through the body in a closed circulatory system. Which chamber of the heart pumps blood to the lungs?

a) Right atrium
b) Left atrium
c) Right ventricle
d) Left ventricle

10. Hormones are chemical messengers that regulate various bodily functions. Which gland is often called the "master gland" because it controls the activity of other endocrine glands?

a) Thyroid gland
b) Adrenal gland
c) Pituitary gland
d) Pancreas

11. The nervous system is responsible for rapid communication and response. Which part of the neuron receives signals from other neurons?
a) Axon
b) Dendrite
c) Cell body
d) Synapse

12.Sexual reproduction involves the fusion of gametes. Which type of cell division produces gametes?
a) Mitosis
b) Meiosis
c) Binary fission
d) Budding

13. DNA replication is a process that ensures genetic information is passed on to daughter cells. Which enzyme unwinds the DNA double helix during replication?
a) DNA polymerase
b) RNA polymerase
c) Helicase
d) Ligase

14. Genetic engineering involves manipulating the genetic material of organisms. Which technique is used to amplify specific DNA sequences?
a) Gel electrophoresis
b) Polymerase chain reaction (PCR)
c) Restriction enzyme digestion
d) DNA sequencing

15. Stem cells are undifferentiated cells that have the potential to develop into various cell types. Which type of stem cell can differentiate into any type of cell in the body?
a) Totipotent stem cells
b) Pluripotent stem cells
c) Multipotent stem cells
d) Unipotent stem cells

16. The process of photosynthesis converts light energy into chemical energy. Which pigment is primarily responsible for absorbing light energy in plants?
a) Chlorophyll a
b) Chlorophyll b
c) Carotenoid
d) Xanthophyll

17. The Calvin cycle is a series of reactions that fix carbon dioxide from the atmosphere. Where does the Calvin cycle take place in plant cells?
a) Mitochondria
b) Chloroplast stroma
c) Cytoplasm
d) Golgi apparatus

18.The Krebs cycle is a series of reactions that break down glucose to release energy. Where does the Krebs cycle take place in eukaryotic cells?
a) Cytoplasm
b) Mitochondria
c) Chloroplast stroma
d) Golgi apparatus

19. The immune system protects the body from pathogens. Which type of white blood cell produces antibodies?
a) Neutrophils
b) Lymphocytes
c) Macrophages
d) Eosinophils

20.The endocrine system regulates various bodily functions through hormones. Which hormone is responsible for regulating blood glucose levels?
a) Insulin
b) Glucagon
c) Adrenaline
d) Thyroxine

CHEMISTRY

1. Which of the following is a chemical change?
a) Melting of ice b) Rusting of iron c) Evaporation of water d) Sublimation of iodine

2. What is the empirical formula of a compound containing 60% carbon, 40% hydrogen?
a) CH2 b) CH3 c) CH4 d) C2H4

3. Which type of bond is formed by the sharing of electron pairs between two atoms?
a) Ionic bond b) Covalent bond c) Metallic bond d) Hydrogen bond

4. Which state of matter has a definite volume but no definite shape?
a) Solid b) Liquid c) Gas d) Plasma

5. Which of the following factors affects the rate of a chemical reaction?
a) Temperature b) Concentration of reactants c) Presence of a catalyst d) All of the above

6. Which of the following is a strong acid?
a) Acetic acid b) Hydrochloric acid c) Carbonic acid d) Boric acid

7. What is the pH of a neutral solution?
a) 0 b) 7 c) 14 d) 10

8. Which of the following is an exothermic process?
a) Photosynthesis b) Evaporation of water c) Combustion of methane d) Melting of ice

9. Which of the following is a transition metal?
a) Sodium b) Calcium c) Iron d) Carbon

10. Which of the following functional groups is present in alcohols?
a) -OH b) -COOH c) -CHO d) -NH2

11. Which of the following is an alkane?
a) C2H2 b) C2H4 c) C2H6 d) C2H5OH

Physics

1. Motion and Force: A ball is thrown vertically upwards with an initial velocity of 20 m/s. What is its velocity after 2 seconds? (Take g = 10 m/s²)

a) 0 m/s
b) 10 m/s upwards
c) 10 m/s downwards
d) 20 m/s downwards

2. Work and Energy: A body of mass 2 kg is lifted vertically upwards through a distance of 2 m. The work done is (Take g = 10 m/s²)

a) 10 J
b) 20 J
c) 40 J
d) 50 J

3. Waves: The distance between two consecutive crests or troughs in a wave is called its

a) Amplitude
b) Wavelength
c) Frequency
d) Time period

4. Thermodynamics: Which of the following is a reversible process?

a) Free expansion of a gas
b) Slow compression of a gas
c) Explosion of a bomb
d) Burning of a candle

5. Electrostatics: Two point charges, each of magnitude Q, are separated by a distance d. The force between them is F. If the distance between them is doubled, the force becomes

a) F/2
b) F/4
c) 2F
d) 4F

6. Current Electricity: The resistance of a conductor depends on its

a) Length
b) Cross-sectional area
c) Resistivity
d) All of the above

7. Electromagnetism: A current-carrying conductor placed in a magnetic field experiences a force. This force is maximum when the angle between the conductor and the magnetic field is

a) 0°
b) 30°
c) 45°
d) 90°

8. Atomic Spectra: The spectrum of hydrogen atom consists of a series of lines. The series in the visible region is called

a) Lyman series
b) Balmer series
c) Paschen series
d) Brackett series

9. Nuclear Physics: In a nuclear fission reaction, a heavy nucleus splits into two lighter nuclei and releases energy. This energy is due to

a) Mass defect
b) Kinetic energy of the products
c) Potential energy of the products
d) All of the above

10.Work, Energy, and Power: A body of mass 10 kg is moving with a velocity of 20 m/s. Its kinetic energy is
a) 100 J
b) 200 J
c) 1000 J
d) 2000 J
11. Electrostatics: The electric field intensity at a point due to a point charge is directly proportional to
a) Distance from the charge
b) Square of the distance from the charge
c) Inverse of the distance from the charge
d) Inverse square of the distance from the charge

QUANTITATIVE REASONING

Arithmetic and Operations

1. Simplify the following expression: 2/3 + 3/4 - 1/6

a) 17/12
b) 11/12
c) 13/12
d) 15/12

2. Find the product of 3/5 and 5/9.

a) 1/3
b) 2/3
c) 1/5
d) 2/5

3. Divide 12 by 2/3.

a) 8
b) 9
c) 10
d) 18

5.What is 20% of 80?

a) 12
b) 14
c) 16
d) 18

4. If a shirt costs Rs. 500 and is discounted by 20%, what is the sale price?

a) Rs. 400
b) Rs. 350
c) Rs. 450
d) Rs. 300

5. A train travels 300 km in 5 hours. What is its average speed?

a) 60 km/h
b) 50 km/h
c) 40 km/h
d) 70 km/h

6. A recipe calls for 2 cups of flour to make 12 cookies. How many cups of flour are needed to make 36 cookies?

a) 4 cups
b) 5 cups
c) 6 cups
d) 7 cups

7. If a car travels 60 miles in 2 hours, how far will it travel in 5 hours at the same speed?

a) 120 miles
b) 150 miles
c) 180 miles
d) 210 miles
Algebra and Functions

8. Simplify the expression: 3x + 2x - 5x

a) 0
b) x
c) 2x
d) -x

9. Solve the equation: 2x + 5 = 15

a) x = 5
b) x = 10
c) x = 7
d) x = 8

10. If f(x) = 2x + 1, what is f(3)?

a) 5
b) 7
c) 9
d) 11

11. What is the slope of the line y = 3x + 2?

a) 1
b) 2
c) 3
d) 4

12. If the equation of a line is y = mx + b, what does "m" represent?

a) y-intercept

b) x-intercept
c) slope
d) constant

13. Simplify the expression: (x^2)^3

a) x^5
b) x^6
c) x^9
d) x^8

Geometry

14. What is the area of a rectangle with length 8 cm and width 5 cm?

a) 30 cm^2
b) 40 cm^2
c) 20 cm^2
d) 50 cm^2

15. What is the perimeter of a square with side length 6 cm?

a) 18 cm
b) 24 cm
c) 30 cm
d) 36 cm

16. The sum of the angles in a triangle is:

a) 180 degrees
b) 270 degrees
c) 360 degrees
d) 90 degrees

17. What is the formula for the area of a circle?

a) A = πr^2
b) A = 2πr
c) A = πr
d) A = πd

Equations

18. Solve the equation: 3x - 7 = 14

a) x = 5
b) x = 7
c) x = 9
d) x = 11

19. Solve the equation: 2(x + 3) = 10

a) x = 1
b) x = 2
c) x = 3
d) x = 4

20. Solve the equation: x^2 - 4 = 0

a) x = 2 or x = -2
b) x = 4 or x = -4
c) x = 1 or x = -1
d) x = 3 or x = -3

21. Solve the equation: 1/x = 2

a) x = 1/2
b) x = 2
c) x = 1
d) x = 4

Statistics

22. The mean of the numbers 5, 7, 9, and 11 is:

a) 6
b) 7
c) 8
d) 9

23. The mode of the numbers 2, 3, 2, 4, 5, 2 is:

a) 2
b) 3
c) 4
d) 5

24. The median of the numbers 1, 3, 5, 7, 9 is:

a) 3
b) 4
c) 5
d) 6

25. If a coin is tossed, what is the probability of getting a head?

a) 1/2
b) 1/4
c) 1/3
d) 2/3

Mental Mathematics

26. What is 15% of 120?

a) 15

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b) 18
c) 21
d) 24

27. If a number is increased by 25%, what is the new number as a percentage of the original number?

a) 100%
b) 125%
c) 150%
d) 175%

28. If a number is decreased by 20%, what is the new number as a percentage of the original number?

a) 80%
b) 75%
c) 70%
d) 65%

29. If a product costs Rs. 100 and is sold for Rs. 120, what is the profit percentage?

a) 10%
b) 15%
c) 20%
d) 25%

30. If a product costs Rs. 100 and is sold for Rs. 80, what is the loss percentage?

a) 10%
b) 15%
c) 20%
d) 25%

31. If a train travels 60 km in 1 hour, how long will it take to travel 300 km at the same speed?

a) 3 hours
b) 4 hours
c) 5 hours
d) 6 hours

32. Calculate 20% OF 150.
a) 20
b)30
c)40
D)50

Verbal Reasoning

Analogies

1. Doctor : Patient :: Teacher : ?

a) Student
b) Book
c) School
d) Pen

2. Hot : Cold :: Day : ?

a) Night
b) Moon
c) Star
d) Sun

3.Rich : Poor :: Wise : ?

a) Clever
b) Foolish
c) Smart
d) Intelligent

4. Car : Road :: Ship : ?

a) Track
b) Sky
c) Sea
d) River

5. Flower : Petal :: Book : ?

a) Cover
b) Page
c) Author
d) Reader

6. **Hammer : Nail :: Knife :** ?

a) Wood
b) Fruit
c) Vegetable
d) Butter

Synonyms

7. Tiny

a) Huge
b) Small
c) Tall
d) Big

8. Ancient

a) Modern
b) New
c) Old

d) Recent

9. Joyful
a) Sad
b) Happy
c) Angry
d) Bored
10 .Courageous
a) Fearful
b) Timid
c) Brave
d) Shy

Antonyms

10. Honest
a) Truthful
b) Sincere
c) Dishonest
d) Genuine

12. Strong
a) Powerful
b) Mighty
c) Weak
d) Robust

13. Kind
a) Gentle
b) Caring
c) Cruel
d) Compassionate

Sentence Completion

14.The old man walked _____ the street.
a) across
b) along
c) through
d) over

15. She was very _____ when she heard the bad news.
a) happy
b) sad
c) excited
d) angry

16. I _____ to school every day.
a) go
b) went
c) gone
d) going

17. She _____ a letter to her friend yesterday.
a) writes
b) wrote
c) written
d) writing

18. He is _____ intelligent boy.
a) a
b) an
c) the
d) no article
19.I like to _____ books in my free time.
a) read
b) write
c) play
d) sing

20.The sun _____ in the east.
a) rises
b) sets
c) shines
d) glows

Comprehension

Read the following passage and answer the questions:

A young boy named Alex loved to explore the woods near his house. One day, he stumbled upon a hidden cave. Inside, he found a treasure chest filled with gold coins. Excited, he took a few coins and left the cave. However, on his way home, he felt a strange sensation. He realized the coins were cursed. They turned into stones, and he was trapped in the woods.

21. What did Alex find in the cave?
a) A treasure chest
b) A magical sword
c) A talking animal
d) A hidden passage

22.What happened to the coins Alex took?
a) They multiplied
b) They disappeared
c) They turned into stones
d) They became worthless

23.Why did Alex feel a strange sensation?
a) He was hungry
b) He was tired
c) He was scared
d) The coins were cursed

24.What was the main idea of the passage?
a) The importance of sharing
b) The dangers of greed

c) The beauty of nature
d) The power of friendship

25.What lesson can we learn from Alex's story?
a) Always be careful what you wish for
b) Honesty is the best policy
c) Never give up on your dreams
d) Greed can lead to misfortune

26. Mission MBBS HEC USAT Pre TEST _____ Helpful .

THE END

For key and Result Follow

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